

EGS3012S project 2026:

Analysing an African extreme weather or climate event

Due date: Monday 19 October 2026, 17:00

Format: A0 virtual Poster (no need to print it!)

Individual submission

Extreme events:

The purpose of this project is to identify and investigate an **extreme weather or climate event** that led to **significant social and/or environmental impacts**. You will need to link the impacts to large (extreme) deviations from expected weather or climate conditions in one or more variables. Please note that floods, fires, high seas and water shortages can result from conditions other than meteorological or climatic factors. Hence, ensure that the event you select can be clearly associated with extreme atmospheric conditions. If you're unsure, feel free to contact Stefaan (s.conradie@uct.ac.za).

Please select one of the following types of extremes:

- Floods linked to extreme rainfall or storm-surge (flash, river or coastal floods)
- Droughts (hydrological, meteorological or agricultural droughts)
- Extreme wind gusts or speeds
- Extreme thunderstorms
- Heat waves
- Extreme cold spells
- Heavy snowfall
- Extreme high seas due to a storm
- Extreme fire weather associated with a destructive wildfire or urban fire

Instructions:

1. Use various sources to identify an impactful extreme weather or climate event **over Africa** which occurred **between January 2016 and July 2026**. Media articles, journal articles, weather data or personal experiences can be referred to. Please ensure that you have **multiple independent sources** to verify that the event occurred, was impactful and resulted from extreme weather or climate anomalies.
2. Post your selected extreme event on: [EGS3012S_2026_project_signup_sheet](#)
3. Indicate the **type of event**, the **time and duration** and the **impact area** you intend to analyse. Ensure that no one else in the class has already chosen the same event

and area as you; each poster needs to **cover a unique event-area-date combination**. Before proceeding, make sure that your event has been **approved by one of the lecturers**. If we have concerns with your event selection, we will be in contact with you to suggest alternatives that may work better. You need to have an event selected by Thursday 20 August 2026, at the latest, or we will assign you with an event not yet taken.

4. Use the sources you found to briefly note some of the **impacts of the extreme event**. Note particularly affected communities, sectors, and/or ecosystems.
5. Use <https://earth.nullschool.net> or an alternative resource (e.g. the [KNMI Climate Explorer](#), [ERA-explorer](#), the [NOAA PSL Weather & Climate Plotting & Analysis Tools](#), the [Climate Reanalyzer](#), [zoom.earth](#), etc.) to **identify the type of weather system or systems** (e.g. cold front, cut-off low, tropical cyclone, high-pressure cell, coastal low with berg winds) that were present during the event. For long-term averages, for example the [KNMI Climate Explorer](#) may be more useful.
6. What **levels in the atmosphere and which variables** are most relevant in identifying and characterising these features? How did these features affect the conditions experienced during the event? What were the **average and most extreme values of the variables you identified above during the event** and how different are these to what is normally experienced? Can you tie these conditions to the **impacts** that occurred?
7. What are the **relevant temporal and spatial scales** for your event?
8. Can you find any evidence that a **significant mode of climate variability** was associated with this event? Or that **climate change** may have played a role?
9. Answer ONE of the following two questions:
 - a. Relate the conditions that occurred during the event to an atmospheric **dynamic or thermodynamic process** you have learnt about during the course.
 - OR
 - b. Use available sources on how **climate change is expected to alter atmospheric dynamics and/or thermodynamics** to hypothesise how climate change may alter future extreme events of this kind. Do you expect that they will happen more or less often, or be more or less intense in future?

The above is intended to guide your work; do not feel you need a detailed answer to every subpoint, but do address each of the 9 points above at least very briefly.

Grading will be done according to the accompanying rubric, with a focus on **demonstrating that you have performed data analysis and visualisation using available tools and on demonstrating critical engagement with multiple relevant sources**.

Make sure data sources are clearly and explicitly indicated for all analyses and figures, to enable reproducibility.

Please ensure that you include an explicit conclusion section!

Presentation of work:

- Posters and slides are the primary format through which scientific findings are communicated to colleagues before submission for peer review.
- The key messages from a poster should be easily and quickly visible, communicating primarily through figures, tables, captions and short bullets. Be creative!
- Try to make your poster visually appealing so someone would want to read it.
- **World limit: 600 (you're welcome to do less)**
- Include your student number and contact details on the poster
- Bullet points are acceptable, full sentences are not required
- Be clear and succinct, avoiding repetition
- Acknowledge all data sources directly and precisely so that anyone reading the poster could access the same data and repeat your analysis exactly
- Ensure that all sources are properly **acknowledged**, cited and listed in a formal reference list/bibliography. Choose a reference style that is compact, but includes at least **author, date, title of the work & DOI/link**. Be consistent in the format style used for all entries. Include inline citations. The reference list need not be in large font and is not included in the word count.

Learning outcomes associated with the project

1. Demonstrate an awareness of the impacts of the weather and climate events studied in the course
2. Demonstrate the ability to independently obtain relevant information about weather and climate conditions through searches of available data.
3. Provide a succinct summary of consolidated information
4. Demonstrate understanding of
 - a. Weather systems and their defining characteristics
 - b. Atmospheric variables and their relationship with weather systems
 - c. The timescales relevant for different event types and the appropriate techniques to assess these
 - d. The 3-D structure of weather systems and the interaction between surface, mid- and upper-level weather features
 - e. The connection between weather systems, weather conditions and their impacts
5. Demonstrate basic data analysis, presentation and visualisation skills
6. To demonstrate an understanding of the link between events at different timescales and the potential impact of long-term changes on short-duration events
7. To engage critically with obtained material, reason and hypothesise based on course learning about novel ideas and demonstrate an understanding of the connection between the theoretical learnings from the course and real-world impacts observed and experienced by course participants and broader society.